Feature Detection in Computer Vision

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Balázs, Paszkál, Vince, Levente, Bence, Antal
Éva, Hajni

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What is feature detection?

- Detects important points in images
- Finds stable structures (corners, edges)
- Used for tracking and motion estimation

goodFeaturesToTrack

- Detects strong corner-like points
- Based on intensity changes in both directions
- Selects stable points for tracking

Output: set of (x, y) coordinates

Corner intuition

A good feature has strong variation in both directions:

$$\lambda_1, \lambda_2 \gg 0 \tag{1}$$

- $\lambda_1, \lambda_2 =$ eigenvalues of gradient matrix
- Large values $\rightarrow$ strong corner
- Flat regions $\rightarrow$ not selected

OpenCV implementation

```python
import cv2

img = cv2.imread("frame.png")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

corners = cv2.goodFeaturesToTrack(
    gray,
    maxCorners=100,
    qualityLevel=0.01,
    minDistance=10
)

for c in corners:
    x, y = c.ravel()
    cv2.circle(img, (int(x), int(y)), 3,
        (0,255,0), -1)
```

Parameters

- **maxCorners** – max number of detected points
- **qualityLevel** – threshold for acceptance
- **minDistance** – minimum spacing between points
- **blockSize** – neighborhood size

Applications

- Optical flow tracking (Lucas-Kanade)
- Camera stabilization
- Robotics perception
- Video analysis